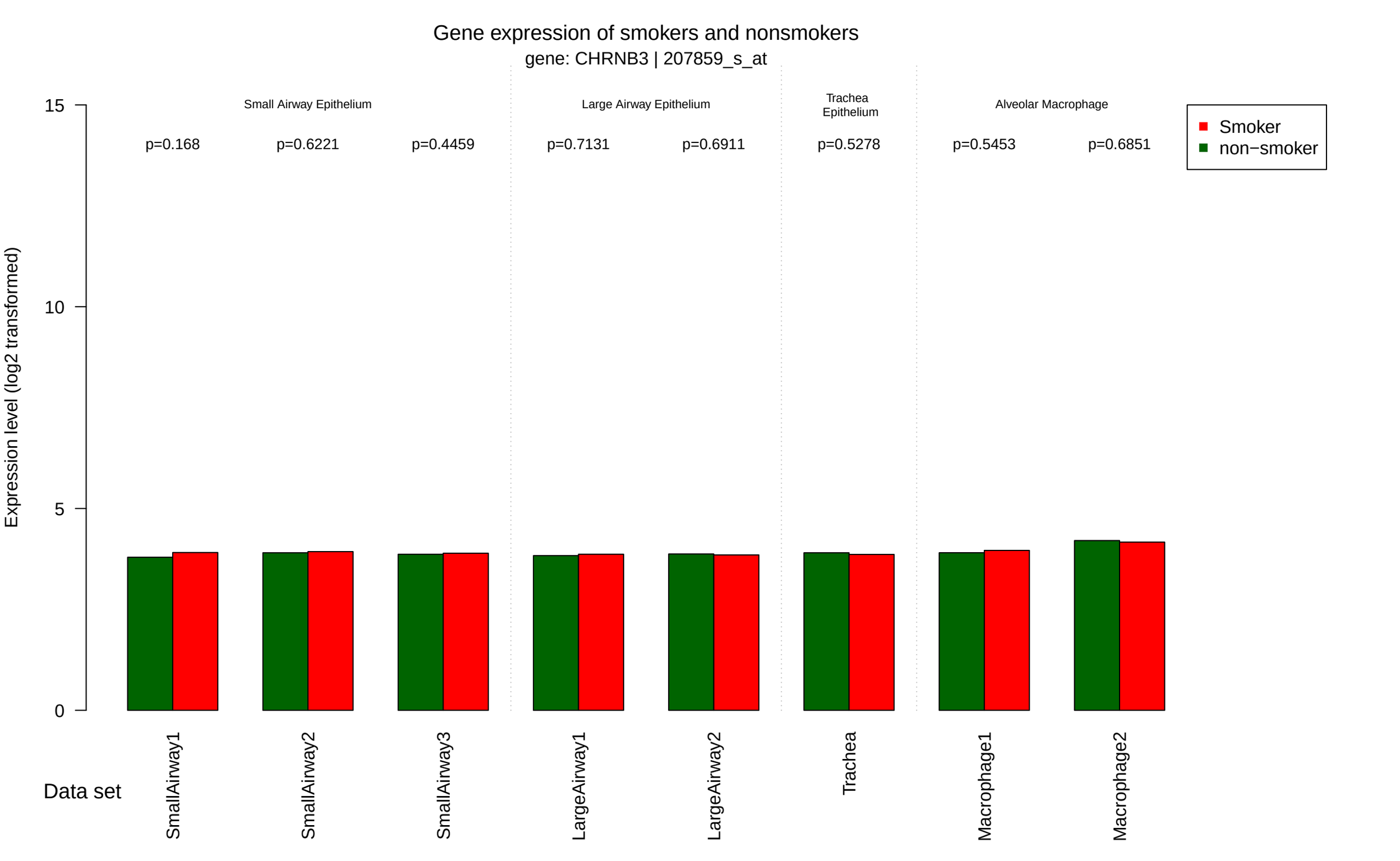
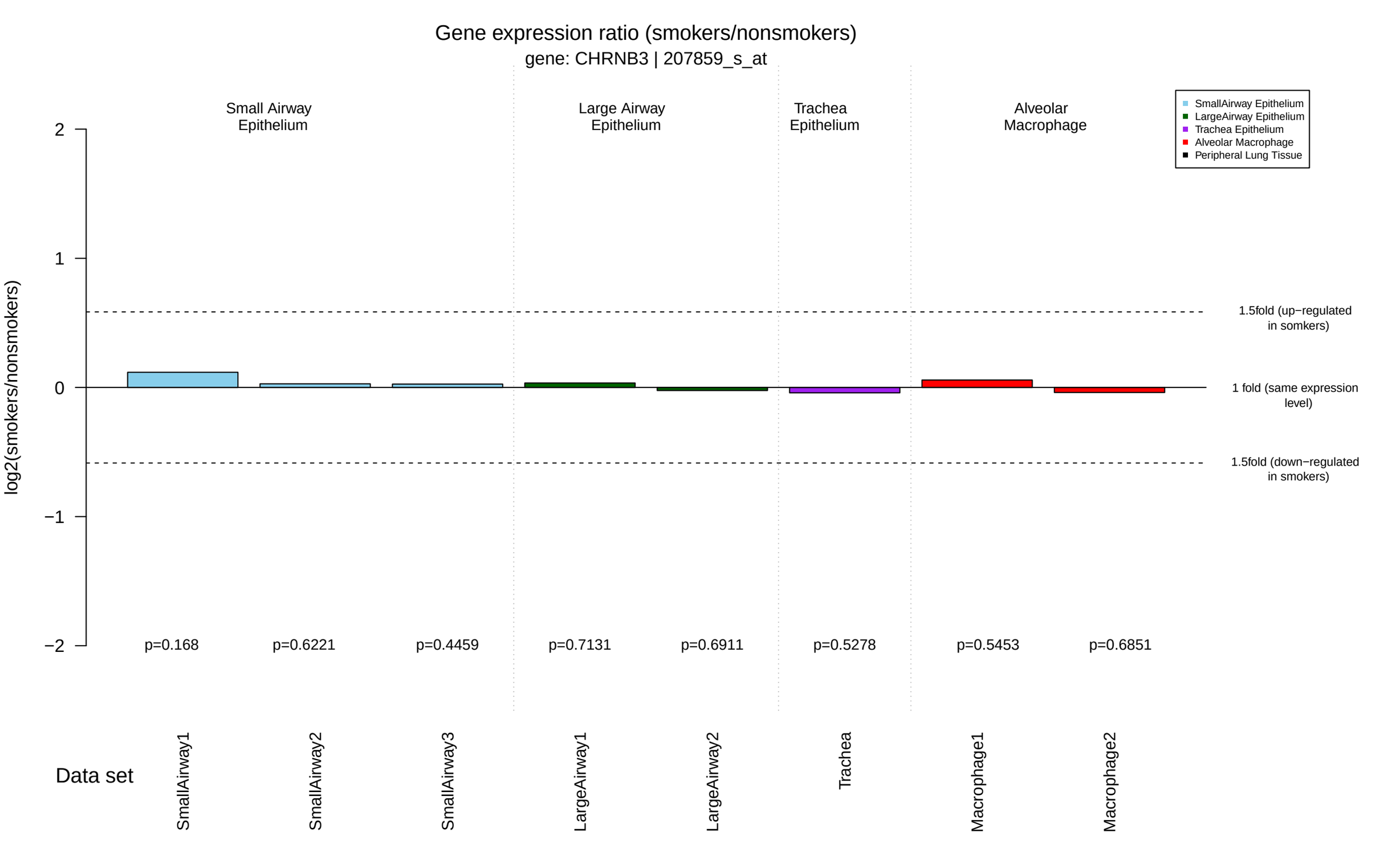
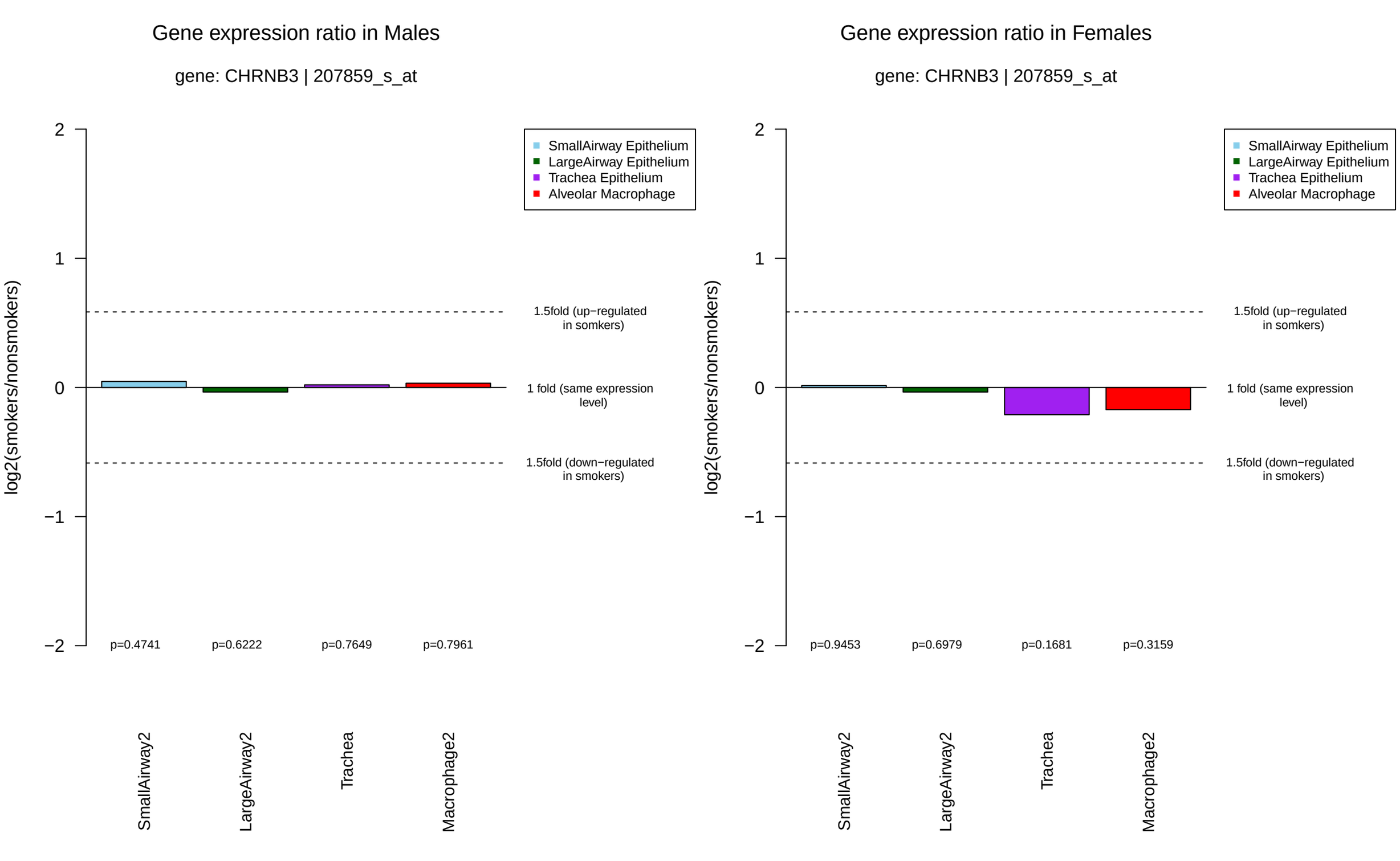
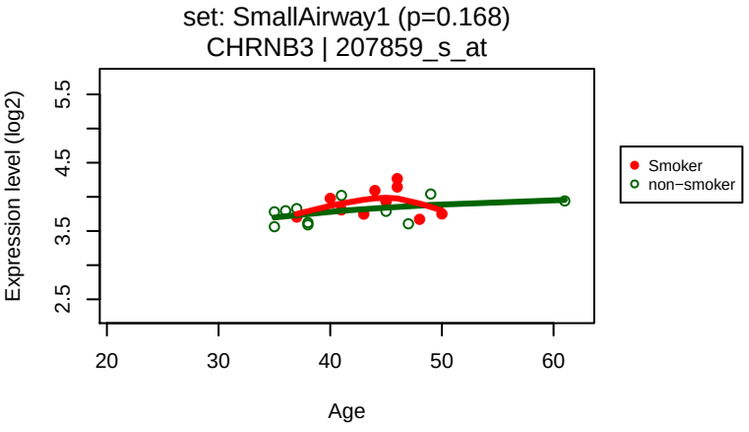
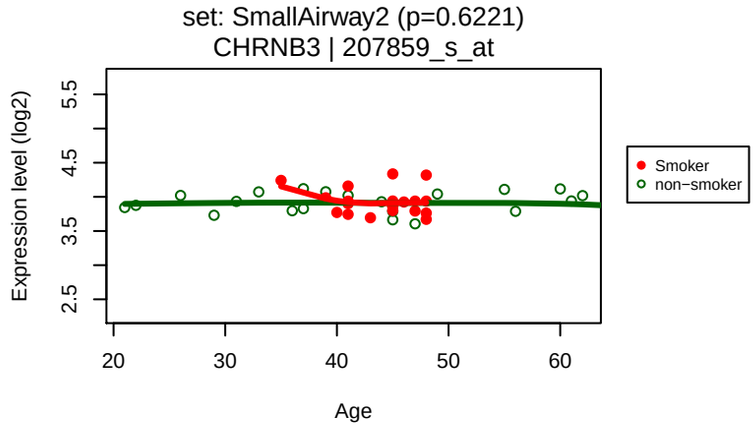
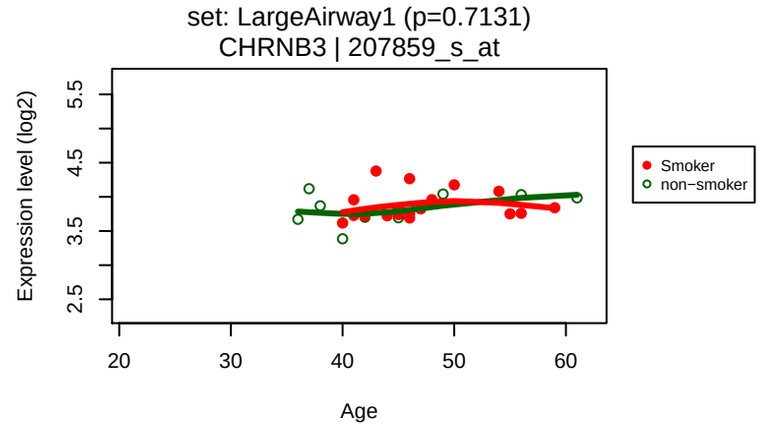

Gene expression of smokers and nonsmokers
gene: CHRNB3 | 207859_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.168
p=0.6221
p=0.4459
p=0.7131
p=0.6911
p=0.5278
p=0.5453
p=0.6851
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

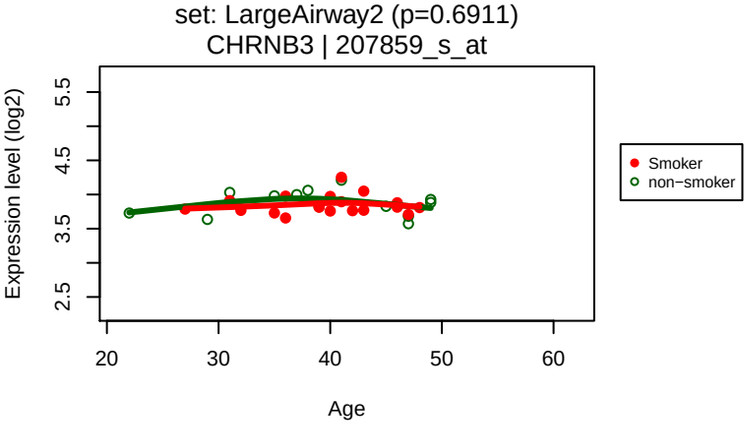

Gene expression ratio (smokers/nonsmokers)
gene: CHRNB3 | 207859_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.168
p=0.6221
p=0.4459
p=0.7131
p=0.6911
p=0.5278
p=0.5453
p=0.6851
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

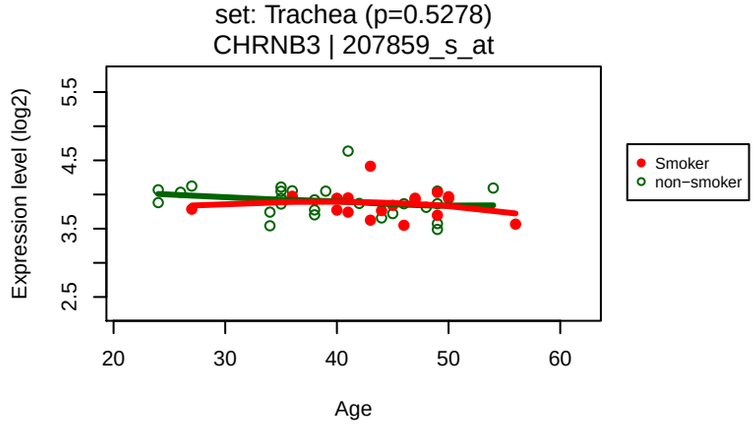

Gene expression ratio in Males
gene: CHRNB3 | 207859_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4741
p=0.6222
p=0.7649
p=0.7961
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRNB3 | 207859_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9453
p=0.6979
p=0.1681
p=0.3159
SmallAirway2
LargeAirway2
Trachea
Macrophage2

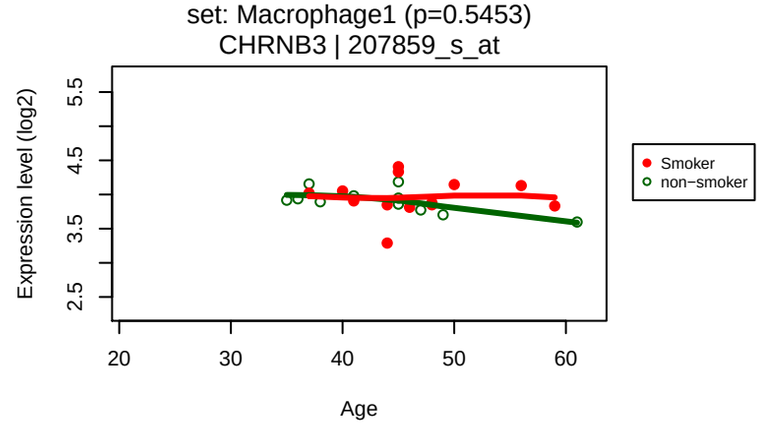

set: SmallAirway1 (p=0.168)
CHRNB3 | 207859_s_at
Expression level (log2)
Age
Smoker
non−smoker

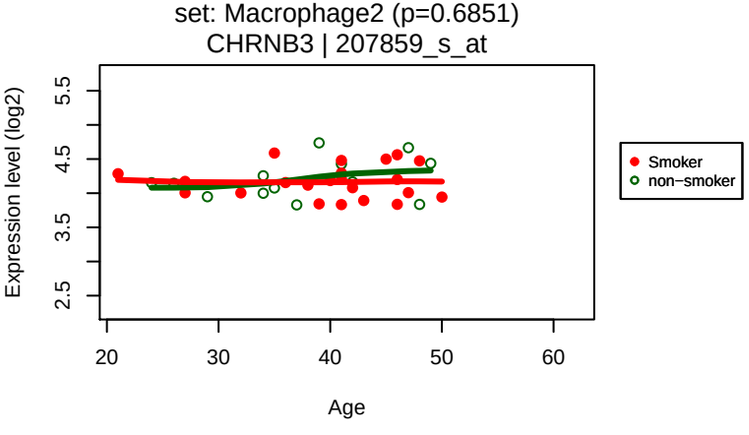

set: SmallAirway2 (p=0.6221)
CHRNB3 | 207859_s_at
Expression level (log2)
Age
Smoker
non−smoker

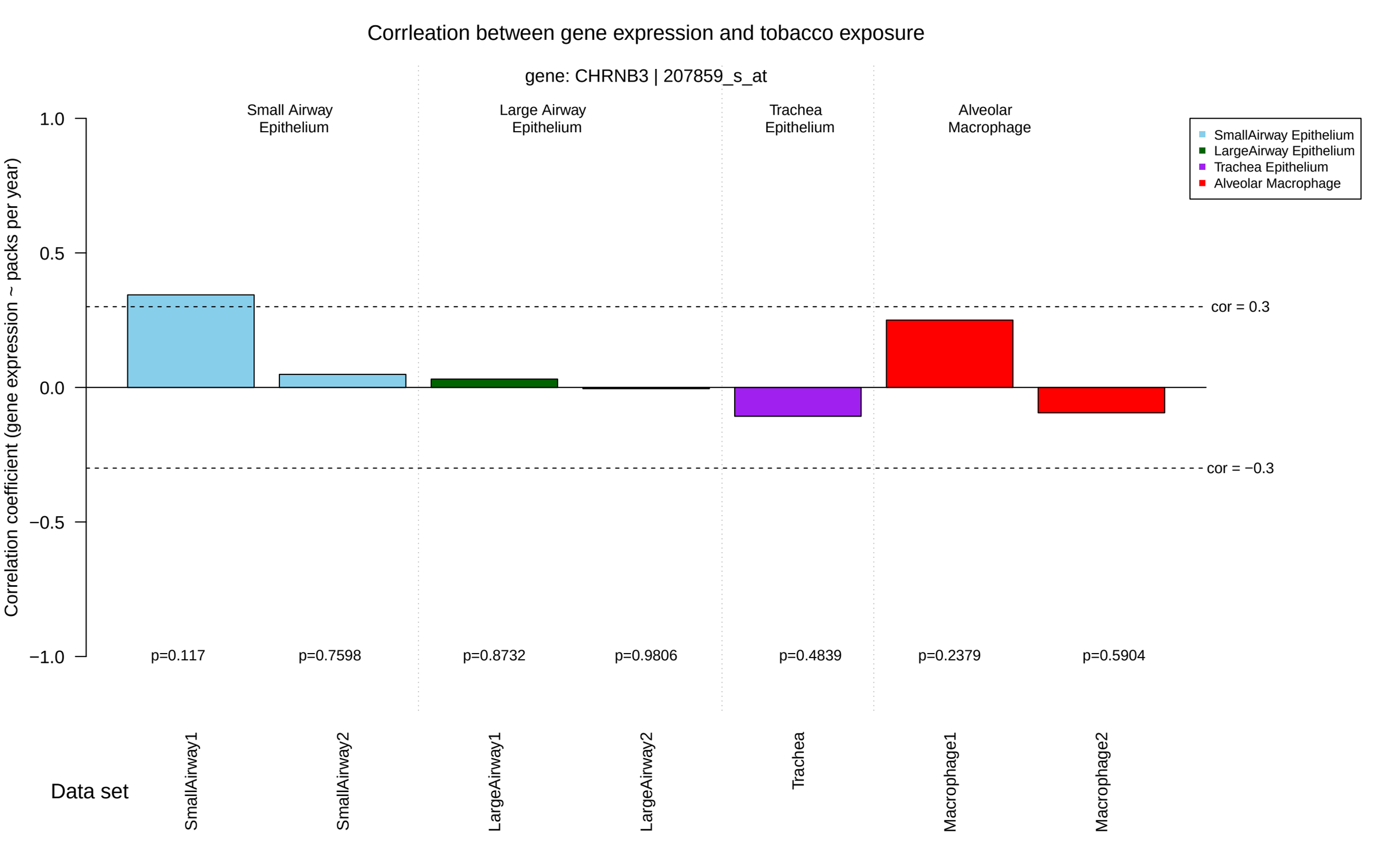

set: LargeAirway1 (p=0.7131)
CHRNB3 | 207859_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6911)
CHRNB3 | 207859_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5278)
CHRNB3 | 207859_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5453)
CHRNB3 | 207859_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6851)
CHRNB3 | 207859_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRNB3 | 207859_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.117
p=0.7598
p=0.8732
p=0.9806
p=0.4839
p=0.2379
p=0.5904
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2